MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 09** | **Time:** 60 Minutes

(QUADRATIC EQUATIONS)

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 13(iii).
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions (VSAQ) of 2 marks each.
4. **Section C** comprises 2 Short Answer questions (SA) of 3 marks each.
5. **Section D** comprises 2 Long Answer questions (LA) of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A (Multiple Choice & Assertion-Reason Questions 6 × 1 = 6 Marks)

Q1. If $2x^2 + px - 15 = 0$ has -5 as a root, and $p(x^2 + x) + k = 0$ has equal roots, then $k =$: [1]

(A) $\frac{7}{4}$ (B) $\frac{4}{7}$ (C) $-\frac{7}{4}$ (D) 7

Q2. If $x = 2$ and $x = 3$ are the roots of the equation $3x^2 - 2kx + 2m = 0$, then: [1]

(A) $k = \frac{15}{2},\ m = 9$ (B) $k = 9,\ m = \frac{15}{2}$ (C) $k = 15,\ m = 18$ (D) $k = -15,\ m = -9$

Q3. The value of k for which $x = -2$ is a root of the equation $3x^2 + 4x + 2k = 0$ is: [1]

(A) 2 (B) -2 (C) 4 (D) -4

Q4. If the sum of the roots of a quadratic equation is 6 and the product of roots is 8, the equation is: [1]

(A) $x^2 - 6x + 8 = 0$ (B) $x^2 + 6x + 8 = 0$ (C) $x^2 - 6x - 8 = 0$ (D) $x^2 + 8x - 6 = 0$

Q5. If the roots of the quadratic equation $kx^2 - 2\sqrt{5}x + 4 = 0$ are equal, then $k =$: [1]

(A) $\frac{5}{4}$ (B) $\frac{4}{5}$ (C) 5 (D) $\frac{\sqrt{5}}{2}$

Q6. Assertion-Reason: [1]

Directions: (A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.

Assertion (A): The sum of the roots of the quadratic equation $3x^2 - 9x + 5 = 0$ is 3.

Reason (R): For any quadratic equation $ax^2 + bx + c = 0$, the sum of roots is given by $-\frac{b}{a}$.

SECTION B (Very Short Answer Questions 2 × 2 = 4 Marks)

Q7. If one root of the quadratic equation $3x^2 - 10x + k = 0$ is reciprocal of the other, find the value of k. [2]

Q8. Solve the following quadratic equation for x by factorization: [2]

$$4x^2 - 4ax + (a^2 - b^2) = 0$$

SECTION C (Short Answer Questions $2 \times 3 = 6$ Marks)

Q9. Solve the following equation for x: [3]

$$\frac{4}{x} - 3 = \frac{5}{2x+3}, \quad x \neq 0, -\frac{3}{2}$$

Q10. If the quadratic equation $(1+m^2)x^2 + 2mcx + (c^2 - a^2) = 0$ has equal roots, prove that: [3]

$$c^2 = a^2(1+m^2)$$

SECTION D (Long Answer / HOTS Questions $2 \times 5 = 10$ Marks)

Q11. In a class test, the sum of Shefali's marks in Mathematics and English is 30. Had she got 2 marks more in Mathematics and 3 marks less in English, the product of her marks would have been 210. Find her marks in the two subjects. [5]

Q12. A piece of cloth costs Rs. 200. If the piece were 5 m longer and each metre of cloth cost Rs. 2 less, the total cost of the piece would have remained unchanged. How long is the piece and what is its original rate per metre? [5]

SECTION E (Applied Case Study Problem $1 \times 4 = 4$ Marks)

Q13. Case Study: Sports Biomechanics & Cricket Bowling Pitch Trajectory

During a high-performance cricket biomechanics camp, the vertical flight trajectory of a delivery bowled by a fast bowler is captured by ultra-high-speed motion cameras. The height h (in metres) of the ball above the pitch surface at a horizontal distance d (in metres) from the bowling release point is modeled by:

$$h(d) = -0.05d^2 + 0.8d + 2.25$$

(i) At what initial release height ($d = 0$) did the bowler release the ball from his hand? [1]

(ii) Set up the quadratic equation in d in standard integer form to find when the ball hits the pitch ($h = 0$). [1]

(iii) Solve the quadratic equation to determine the horizontal distance d at which the ball pitches. [2]

OR

Determine the maximum height attained by the cricket ball above the pitch during its flight.

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★